OXFORD

Briefings in Bioinformatics, 2025, **26(4)**, bbaf425
https://doi.org/10.1093/bib/bbaf425
Problem Solving Protocol

Reg2ST: recognizing potential patterns from gene expression for spatial transcriptomics prediction

Xu Wang[1], Yaqiu Wang[1], Xuan Wang[1,2], Yadong Wang[3,4], Junyi Li[1,2,4,*]

[1]School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, Guangdong, China
[2]Guangdong Provincial Key Laboratory of Novel Security Intelligence Technologies, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, Guangdong, China
[3]Faculty of Computing, Center for Bioinformatics, Harbin Institute of Technology, Harbin 150001, Heilongjiang, China
[4]Key Laboratory of Biological Bigdata, Ministry of Education, Harbin Institute of Technology, Harbin 150001, Heilongjiang, China

*Corresponding author. E-mail: lijunyi@hit.edu.cn

Abstract

Spatial transcriptomics is able to acquire cellular gene expression while retaining spatial location. It is often accompanied by matched hematoxylin and eosin-stained histology whole-slide images. This retention of spatial information is critical for studying key issues in cell biology, developmental biology, neurobiology, and tumor biology. However, conventional sequencing technologies are costly and time-consuming, limiting the development of spatial transcriptomics research. Recently, deep learning methods have been widely applied to spatial transcriptomics prediction, but there are some problems in existing methods. To tackle the problem, we develop Reg2ST, a deep learning model to learn potential patterns in gene expression and apply them for spatial transcriptomics prediction. Reg2ST treats spatial transcriptomics and histology as different expressions of the same data. Contrastive learning is used to minimize the distance between them. Then image features are used to predict gene features, which aligns histology images with spatial transcriptomics. Reg2ST uses a novel way to capture relationships among spots instead of K-Nearest-Neighbors. Evaluations of Pearson correlation coefficient, statistical tests, computational efficiency using human breast cancer, and cutaneous squamous cell carcinoma datasets demonstrate the superior performance of Reg2ST for spatial gene expression prediction.

Keywords: spatial trainscriptomics prediction; histology images; contrastive learning; transformer

Introduction

Understanding the structure of cells and how they affect functions is important for life science researchers. The development of spatial transcriptomics technology has made it possible. In recent years, more and more spatial transcriptomics methods and analysis platforms have emerged, such as ST-Net [1], HisToGene [2], Hist2ST [3], THItoGene [4], HGGEP([5]), OmiCLIP [6], STExplore [7], and stCluser [8]. Spatial transcriptomics can obtain the expression of cellular genes while preserving their spatial information, which is usually accompanied by matched hematoxylin and eosin-stained histology whole-slide images (WSI) [9–11]. According to different spatial transcriptomics technologies, the sequencing expression of each spot usually represents two to dozens of cells [12]. As unique types of cells and specific environments are highly related to biological and biomedical activities, retaining the spatial information is crucial.

Although spatial transcriptomics plays a key role in analyzing the complicated relationship between cells and their microenvironment, its high cost and time-consuming operations hinder its application in large-scale research. To overcome these limitations, researchers have applied deep learning methods to predict spatial transcriptomics from histology images. For example, Schmauch *et al.* [13] successfully verify the idea by predicting many gene expression profiles from WSIs. They develop a method called HE2RNA to accurately capture the microstructural information in WSI and reveal the key tumor areas for developing specific cancer types. Compared with expensive spatial transcriptomics sequencing, these histology image-based prediction methods have broader applications. They combine abundant visual features in histological and biomedical images and simulate the spatial distribution of spatial transcriptomics through machine and deep learning methods. With the continuous improvement of deep learning and histology analysis methods, models similar to HE2RNA are expected to play a greater role in spatial transcriptomics prediction, opening up new directions for the integration of genomics and pathology.

After that, more and more deep learning methods have emerged to predict spatial transcriptomics. ST-Net uses cell morphology with gene expression to achieve spatial transcriptomics prediction from histology images. However, it fails to fully utilize the spatial location of spots. HisToGene uses an improved Vision Transformer [14] to extract features while embedding the spatial information of spots into image features for spatial transcriptomics prediction. However, HisToGene fails to fully

Received: March 17, 2025. **Revised:** June 10, 2025. **Accepted:** July 15, 2025
© The Author(s) 2025. Published by Oxford University Press.
This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

exploit the global and local relationships between spots. Hist2ST achieves better spatial transcriptomics prediction accuracy by cropping images into patches and sequentially applying Convimixer, Transformer [15], and graph neural network (GNN) [16, 17] modules for prediction. The first two modules aim to capture visual features and their intrinsic connections within patches, while GNN perceives global and local relationships between spots. Similarly, THItoGene uses capsule networks [18] to extract visual features from histology images, then integrates spatial location with image features through a graph attention network [19] to fully understand the characteristics of spots for spatial transcriptomics prediction. HGGEP uses a gradient enhancement module for more accurate image feature extraction. Then different scale features are fed into hypergraph to capture associations among them. In this way, HGGEP can learn both within features and connections among patches. OmiCLIP aligns histology images and spatial gene expression into a shared embedding space via contrastive learning. It consists of two main components: an image encoder for WSI and a gene expression encoder for text sentences, which are generated by combining the top 50 expressed genes per spot into sentences. OmiCLIP is trained on 2.2 million tissue patches from 1007 samples across 32 organs. Built upon OmiCLIP, Chen *et al*. [6] further develop Loki platform, capable of predicting gene expression directly from histology images. Experimental results show that Loki achieves state-of-the-art performance on various spatial transcriptomics datasets. The gradual development of these methods provides the possibility of predicting spatial transcriptomics at lower cost and higher efficiency and also lays the foundation for the future application of deep learning technology in spatial transcriptomics.

However, there are three main problems with those methods. First, they simply use histology images for prediction, which fails to include complex knowledge from gene expression. There are plenty of potential patterns underneath gene expression. Identifying these patterns and applying them to images will improve the prediction accuracy. Secondly, histology images contain complex cell information and histological features. Yet existing methods lack sufficient flexibility and depth to capture these complicated structures, making it harder for image feature extraction. More accurate feature extraction methods are needed. Finally, although existing methods have utilized the spatial location of spots to a certain extent, they use the K-Nearest-Neighbors (KNN) algorithm to construct a graph and describe relations between spots, in which spots close to each other will become neighbors. However, proximity in physical distance does not mean complex relations. There are deeper relationships between spots such as hierarchical structure and similar gene expression. More suitable graph construction methods are needed.

Therefore, to face the above challenges, we propose Reg2ST, which aims to effectively recognize patterns from gene expression and apply them to predict. By doing this, histology images and spatial transcriptomics are regarded as different expressions of the same spot in different modalities. Then we use contrastive study to learn the potential relationship between spatial transcriptomics and histology images and align histology images to spatial transcriptomics. Finally the trained model is used to embed histology images and predict spatial transcriptomics.

Reg2ST's architecture is shown in Fig. 1, which contains three modules: feature extraction module, feature fusion module, and dynamic GNN module. In the feature extraction module, histology patches and gene expression are input into the image encoder and gene encoder to extract image and gene features, respectively. By minimizing the cosine similarity between these two features, Reg2ST can narrow the distance between image and gene features.

In the feature fusion module, image features are first used to generate predicted gene features through a prediction layer. Then, the coordinates of spots are embedded into image features to learn spatial relationships among spots. After that, enhanced image features and predicted gene features are input into a cross-attention layer to form richer fusion features.

Finally, predicted gene features, image features, and fused features are input into the dynamic GNN module. The GNN further improves the features by integrating local and global relationships among spots. After obtaining high-quality features, the gene expression is accurately predicted using the zero-inflated negative binomial (ZINB) distribution model [20]. Comprehensive benchmark tests are conducted on the spatial transcriptomics generated by different tumors. Results show that our model performs well and provides an innovative solution for efficient spatial transcriptomics prediction from histology images.

Materials and methods

Datasets and preprocessing

In this study, the same spatial transcriptomics datasets as in the previous study are used, including two datasets: the human HER2-positive breast cancer dataset (HER2+) [21] and the human cutaneous squamous cell carcinoma (cSCC) dataset [22]. Both of them contain histology images, gene expression, and spot coordinates. The HER2+ dataset and cSCC dataset consist of 36 tissue sections from eight patients and 12 tissue sections from four patients, respectively.

Due to problems like noise, batch effects from raw spatial transcriptomics, it is essential to preprocess histology images and gene expression. Histology images are cropped into patches of 112 × 112 pixels. As for gene expression, the top 1000 highly variable genes are selected for each section, which can better reflect the heterogeneity between tissues. At the same time, genes with <1000 spots in all sections are excluded to remove the noise of low-expression genes. To standardize the gene count value of each spot, the count of each gene is divided by the total count of the point and multiplied by a scale factor (1000 000). Subsequently, the normalized count is converted to a logarithmic scale using the natural logarithm transformation formula. After the above preprocessing, the HER2+ dataset finally retained 9612 spots and 785 genes, and the cSCC dataset retained 6630 spots and 171 genes.

Prediction strategy

Before introducing our method, it is necessary to explain our prediction strategy, which is the main idea in our study. The objective of Reg2ST is to recognize underlying patterns in gene expression, then apply them to associated histology images for spatial transcriptomics prediction. So histology images and spatial transcriptomics are treated as different expressions of the same spot. Contrastive study is used to learn the potential relationship between spatial transcriptomics and histology images. After that, a unique strategy is developed for prediction: predict gene features using histology image features, then predict spatial transcriptomics using predicted gene features.

There are two reasons. First, spatial transcriptomics contains much richer information than single histology image such as spot heterogeneity and gene interactions. By predicting gene features, Reg2ST can further mine and utilize the information between

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

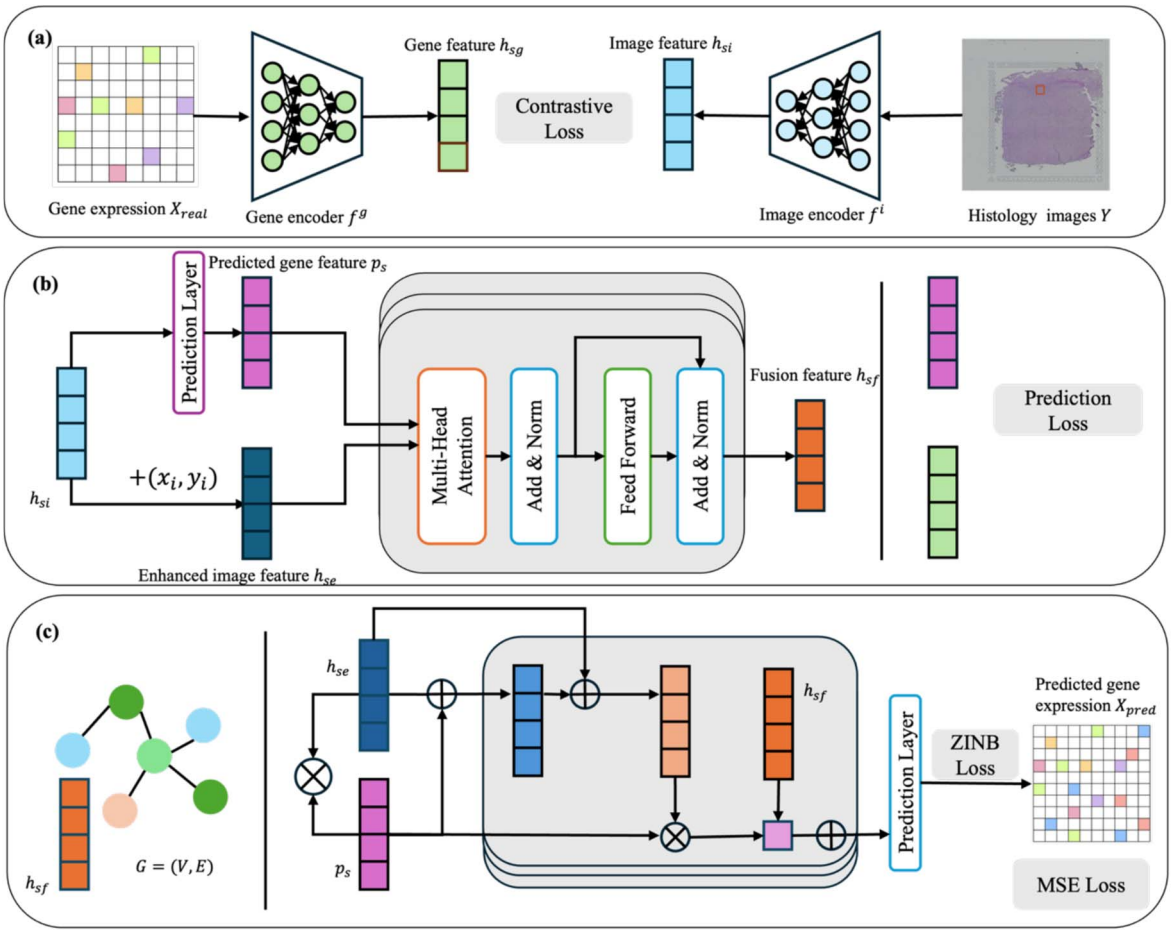

Figure 1. The pipeline of Reg2ST. (a) The feature extraction module: gene expressions and histology images are input into the gene encoder and image encoder to get gene features and image features, respectively. (b) The feature fusion module: first image features are used to predict gene features through a prediction layer, while spatial coordinates of spots are embedded and integrated into image features. Then predicted gene features and enhanced image features are input to a cross-attention layer and obtain fusion features. (c) The dynamic GNN module: predicted gene features, enhanced image features, and fusion features are input into the dynamic network module to perceive the local and global relationships between spots. Spatial transcriptomics are predicted by following by ZINB distribution.

spots, thereby providing more accurate support for subsequent spatial transcriptomics predictions. The results of ablation studies also prove that usage of this strategy significantly improves prediction accuracy, indicating that gene features play a key role in Reg2ST.

Secondly, predicted gene features and enhanced image features are input into the cross-attention layer. Through the cross-attention mechanism, enhanced image features and predicted gene features interact deeply. Valuable hidden patterns are extracted and integrated to fusion features. This process not only enhances the associations between spatial transcriptomics and histology images, but also guides Reg2ST to focus more on the most critical parts in prediction. For example, positional coordinates in spatial transcriptomics and highly expressed regions of specific genes can help the model locate the corresponding pathological structure and cell morphology in the image, thereby optimizing the final prediction performance.

In summary, by introducing gene features in the middle layer, the two-staged prediction strategy not only effectively improves the prediction accuracy of the model, but also alleviates the gap between image features and gene features to a certain extent.

Overview of Reg2ST

Reg2ST recognizes the hidden patterns in gene expression and applies them to histology images to predict spatial transcriptomics. It contains a feature extraction module, a feature fusion module, and a dynamic GNN module. The three modules work together to predict spatial transcriptomics.

The feature extraction module contains two encoders to capture key features of gene expression and histology images, respectively. In the feature fusion module, image features are first input into the prediction layer to obtain predicted gene features. Next, the coordinates of each spot are embedded into the image features to form enhanced image features. Subsequently, predicted gene features and enhanced image features are input into a cross-attention layer for feature fusion. The cross-attention layer captures the correlation and interaction between different features to generate fusion features of each spot. Fused features can simultaneously reflect spatial patterns of gene expression and the structural information of image features.

In the dynamic GNN module, a graph is constructed based on the predicted gene features and enhanced image features, which are two crucial attributes of spots and reflect their heterogeneity. Specifically, spots are organized into a dynamic graph using predicted gene features and image features. Compared with

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

traditional KNN-based models, the graph in Reg2ST dynamically changes during the training process. This enables Reg2ST to capture biological relationships between spots better. Subsequently, fusion features of spots are input into the dynamic GNN as the initial features to perceive global and local relationships between spots, thereby further optimizing the feature representation. After completing feature extraction and optimization, we introduce the ZINB distribution model, which is suitable for processing high-dimensional and sparse gene expression, thereby achieving accurate prediction of gene expression.

Feature extraction module

Using pretrained large models for histology image feature extraction has become a widely adopted trend because of batch effect and noise. Directly training models on small-scale datasets may also lead to overfitting, while pretrained models can effectively alleviate this problem.

In this study, a pretrained large model Phikon-v2 [23] is selected as the image feature extractor. Phikon-v2 is a large-scale public feature extraction model specifically designed for biomarker prediction.

Histology images have been cropped into 112 × 112 pixel patches and input into Phikon-v2 first for feature extraction. Then image encoder f^i, a multilayer perceptron (MLP), is applied to map features to latent space. The process is illustrated in Equation (1), where $y_s \in R^{C*H*W}$ denotes as the patch of spot s, and h_{si} denotes the image feature of spot s.

$$h_{si} = f^i(Phikonv2(y_s)) \tag{1}$$

For gene expression, we also use an MLP as the gene encoder. The process is demonstrated in Equation (2), where $X \in R^{m*n}$ denotes as the gene expression with m spots and n genes, f^g denotes as the gene encoder, and h_{sg} denotes the gene feature of spot s.

$$h_{sg} = f^g(x_s) \tag{2}$$

Feature fusion module

The feature fusion module aims to generate fused feature for each spot by capturing the correlations and interactions between spatial transcriptomics and histology images. Fused features can reflect both the potential patterns of spatial transcriptomics and the complex structure of image features.

For starters, image features are used to predict gene features by a prediction layer, which contains two fully connected layers, as Equation (3) demonstrated, where p_s denotes predicted gene feature of spot s.

$$p_s = PredLayer(h_{si}) \tag{3}$$

Secondly, to fully use spatial information of spots, the coordinates of each spot are embedded and integrated into associated image features. The process is demonstrated as Equation (4), where e_x and e_y denote embeddings of x and y coordinates, respectively, and h_{se} denotes the enhanced feature of spot s.

$$h_{se} = h_{si} + e_x + e_y \tag{4}$$

Then the fused features are obtained through a cross-attention layer. Through this, deep associations and interactions can be established between image features and predicted gene features. Fused features are computed through Equation (5) to Equation (8), where W_Q, W_K, and W_V denote learnable parameter matrices for query, key, and value, and Q, K, and V denote query, key, and value vectors, respectively, and d is the dimension of latent vector.

$$K = W_k * p_s \tag{5}$$

$$V = W_v * p_s \tag{6}$$

$$Q = W_Q * h_{se} \tag{7}$$

$$CrossAttention(p_s, h_{se}) = Softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{8}$$

Dynamic GNN module

Conventional spatial transcriptomics prediction usually uses KNN to construct a graph. However, it fails to capture biological associations such as spot heterogeneity and tissue structure information. Spatial transcriptomics has unique multidimensional characteristics, requiring a more flexible and diverse approach to construct graphs, which will fully utilize the complex relationships between spots. For a spot, its tissue structure and gene expression are of great importance, which define the functional status and determine its association with other spots. Therefore, in our method, we use enhanced image features and predicted gene features to construct a graph.

We follow the modified WiKG [24] algorithm. It is a novel dynamic graph representation method that can effectively capture the complex associations between nodes through dynamic relationship modeling and knowledge-aware attention mechanisms. The algorithm performs well in dynamic graph modeling and generates a more comprehensive node representation for each spot, providing inspiration and technical support.

Specifically, a graph is constructed based on enhanced image features and predicted gene features of each spot. The dot product between each pair of spots is to quantify the similarity between the two features and then normalized. The normalized similarity value represents the strength of association between spots, as shown in equation (9), where $\omega_{(s,t)}$ denotes the similarity score between h_{se} and p_t. Through this, spots similar to s can be defined as its neighbors.

$$\omega_{(s,t)} = \frac{h_{se}^T p_t}{\sum_{t=1}^{N}(h_{se}^T p_t)} \tag{9}$$

For each spot s, the top k spots with the highest similarity scores are selected as the neighbors of spot s, denoted as $N(s)$. The process can be illustrated in Equation (10), where V denotes the spot set, and $|V| = m$, $|N(s)| = k$.

$$N(s) = \left\{t \in V : \omega_{s,t} \in TopK\{\omega_{s,t}; t = 1, 2, \ldots, N\}\right\} \tag{10}$$

Then a directed graph is constructed based on the image features and spatial transcriptomics features of spots, and an embedding is assigned to each edge, as shown in Equation (11), where $r_{s,t}$ represents the edge embedding from spot t to spot s. At the beginning of training, the predicted gene features obtained by Reg2ST can neither be fully aligned with the real gene features, nor reflect the interactions between genes well. While the predicted gene features gradually aligned with real gene features, the

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

relationships between spots are gradually corrected.

$$r_{s,t} = \omega_{s,t} h_{se} + (1 - \omega_{s,t}) p_t, \ for\ every\ t \in N(s) \tag{11}$$

Through the above operations, the relationship between spots is described as a dynamic graph $G = (V, E, F, R)$, where V denotes the set of nodes, E denotes the set of edges, F denotes the set of enhanced image features and predicted spatial transcriptomics features, and R represents the set of directed edge embeddings.

After that, a knowledge-based attention layer is used to perceive global and local relationships. For spot s, its first-order connectivity structure is calculated using Equation (12) to Equation (14), where $\pi(h, r, p)$ represents a weighting factor to leverage how much information each spot propagates to neighbors.

$$h_{N(s)} = \sum_{t \in N(s)} \pi(h_{se}, r_{s,t}, p_t) p_t \tag{12}$$

$$u(h_{se}, r_{s,t}, p_t) = p_t^T \tanh(h_{se} + r_{s,t}) \tag{13}$$

$$\pi(h_{se}, r_{s,t}, p_t) = \frac{\exp\{u(h_{se}, r_{s,t}, p_t)\}}{\sum_{t \in N(s)} \exp\{u(h_{se}, r_{s,t}, p_t)\}} \tag{14}$$

Subsequently, a dual-interaction mechanism is used to generate representations for each spot, which is computed using Equation (15), where h_{sf} denotes the fusion features of spot s from feature fusion module.

$$h_{sd} = \sigma_1(W_1(h_{sf} + h_{N(s)})) + \sigma_2(W_2(h_{sf} \cdot h_{N(s)})) \tag{15}$$

Finally, an MLP is used to generate spatial transcriptomics prediction results following ZINB distribution rules. The process is illustrated in Equation (16) to Equation (18), where $drop, dist,\ and\ mean$ are parameters of ZINB distribution, and S_s is the ratio of the total spot count to the median.

$$drop = Sigmod(W_\pi h_{sd}) \tag{16}$$

$$disp = \exp(W_\theta h_{sd}) \tag{17}$$

$$mean = diag(S_s) * \exp(W_\mu h_{sd}) \tag{18}$$

Loss function

The overall loss function of Reg2ST consists of four parts: contrastive learning loss, feature prediction loss, ZINB distribution loss, and reconstruction loss.

To learn potential patterns in spatial transcriptomics and align histology images to spatial transcriptomics, we design contrastive learning loss and feature prediction loss. Contrastive learning loss is commonly used to measure the similarity between pairs. Specifically, spatial transcriptomics contains rich gene expression, while histology images record cell morphology and tissue structure characteristics. Although these two types of data come from different modalities, they are inherently related in biology. Therefore, the gene features and image features from the same spot are set as a positive pair. Contrastive learning loss is computed by Equation (19) to Equation (22), where $l(s, t)$ denotes the contrastive learning loss between a positive pair, and τ denotes the temperature parameter.

$$l(s, t) = -\log \frac{\exp(sim(s, t)/\tau))}{\sum_{k=1}^{m} \exp(sim(s, k)/\tau)} \tag{19}$$

$$sim(s, t) = \frac{s^T t}{\|s\| \, \|t\|} \tag{20}$$

$$L_{image} = \frac{1}{m} \sum_{s=1}^{m} l(h_{si}, h_{sg}), L_{gene} = \frac{1}{m} \sum_{s=1}^{m} l(h_{sg}, h_{si}) \tag{21}$$

$$L_{con} = \frac{L_{image} + L_{gene}}{2} \tag{22}$$

By optimizing the contrast loss function, the model can more accurately learn the key information related to spatial transcriptomics hidden in the image, to better model the deep relationship between image features and spatial transcriptomics features.

The mean square error (MSE) between predicted gene features and real gene features is defined as feature prediction loss to measure the distance between them. Compared with simply predicting spatial transcriptomics, feature prediction loss based on feature alignment can significantly improve the accuracy of prediction results. It can be computed following Equation (23), where p_s represents predicted gene features and h_{sg} represents real gene features from the gene encoder.

$$L_{pred} = \sum \|h_{sg} - p_s\|_2^2 \tag{23}$$

In spatial transcriptomics prediction methods, direct use of MSE loss may perform poorly due to coefficient row and discreteness of the data. By introducing the ZINB distribution to model zero values, the distribution characteristics of highly expressed genes are better described. The accuracy of spatial transcriptomics prediction is also improved. The process is illustrated in Equation (24) to Equation (26), where $drop, dist,\ and\ mean$ are parameters mentioned above, written as π, θ, and μ, respectively.

$$NB(X|\mu, \theta) = \frac{\Gamma(X + \theta)}{X! \, \Gamma(\theta)} \left(\frac{\theta}{\theta + \mu}\right)^\theta \left(\frac{\mu}{\theta + \mu}\right)^X \tag{24}$$

$$ZINB(X|\pi, \mu, \theta) = \pi \varphi(X) + (1 - \pi) NB(X|\mu, \theta) \tag{25}$$

$$L_{ZINB} = \sum -\log\left(ZINB(X|\pi, \mu, \theta)\right) \tag{26}$$

Last but not least, we also apply MSE to measure the accuracy of the prediction results as usual, demonstrated in Equation (27), where X_{real} and X_{pred} denote real and predicted spatial transcriptomics, respectively. And the overall loss function is like Equation (28). In this setting, we set $\lambda_1 = 0.5$ and $\lambda_2 = 0.25$ for all datasets.

$$L_{recon} = \sum \|X_{real} - X_{pred}\|_2^2 \tag{27}$$

$$L = \lambda_1 L_{con} + L_{pred} + \lambda_2 L_{ZINB} + L_{recon} \tag{28}$$

Experiment and results

Evaluated metrics and criteria

The performance of Reg2ST with other baseline methods is evaluated and compared using Pearson correlation coefficient (PCC) evaluation metrics. The value of PCC ranges between 1 and −1, which measures the degree of correlation between predicted spatial transcriptomics and real spatial transcriptomics. It is shown in Equation (29), where $Cov()$ denotes the covariance, X_{true} and X_{pred} denote the real and predicted spatial transcriptomics, respectively, and $Var()$ denotes standard deviations.

$$PCC = \frac{Cov(X_{true}, X_{pred})}{Var(X_{true}) Var(X_{pred})} \tag{29}$$

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

Table 1. Hyperparameter settings of Reg2ST

Hyperparameters	
Feature embedding	256
Number of neighbors	6
λ_1	0.5
λ_2	0.25
Training epochs	400
Learning rate	1e-4

Table 2. Wilcoxon signed-rank test results on HER2+ dataset

Model	Median diff	Wilcoxon *P*-value	Significance ($\alpha = 0.05$)
HisToGene	0.0787	0.0000	Yes
Hist2ST	0.181	0.0003	Yes
THItoGene	0.275	0.0000	Yes
HGGEP	0.0077	0.0002	Yes
Loki	0.1481	0.0000	Yes

Table 3. Wilcoxon signed-rank test results on cSCC dataset

Model	Median Diff	Wilcoxon *P*-value	Significance ($\alpha = 0.05$)
HisToGene	0.1081	0.0002	Yes
Hist2ST	0.101	0.0212	No
THItoGene	0.0093	0.0386	No
HGGEP	0.0704	0.0002	Yes
Loki	0.1686	0.0002	Yes

Table 4. Computational efficiency results on models

Model	Estimated parameters size (MB)	Training tme (s)	CPU usage (MB)	Maximu m GPU usage (MB)
HisToGene	426.593	789.11 (100 epochs)	36 294.87	2361.48
Hist2ST	303.159	2713.24 (400 epochs)	39 267.27	8910.57
THItoGene	254.347	1371.71 (400 epochs)	38 061.93	2266.06
HGGEP	465.460	12 759.37 (400 epochs)	39 954.39	22 970.56
Reg2ST	44.247	747.98 (400 epochs)	35 485.07	762.05

Besides, the Wilcoxon signed-rank test is performed on the PCC between predicted gene expression and true values for the HER2+ and cSCC datasets, to test whether the Reg2ST is statistically significant comparing with baselines. Specifically, for each slice of each dataset, the PCC differences between Reg2ST and baselines are calculated separately. Then a test is performed to determine whether the median of these differences was significantly greater than zero.

Experiment

To validate prediction performance, four published state-of-the-art spatial transcriptomics prediction methods in recent years: HisToGene, Hist2ST, THItoGene, and HGGEP are take into comparison. All models are evaluated on the HER2+ (containing 32 tissue sections) and cSCC (containing 12 tissue sections) spatial transcriptomics datasets by leave-one-out cross-validation. Detailed hyperparameter settings are shown in Table 1. In addition, Loki is incorporated as a baseline method to investigate its performance on our datasets. Specifically, a leave-one-out evaluation strategy is adopted in which one tissue section is selected as the test sample, and the remaining slides serve as reference data.

PCC results of HER2+ and cSCC datasets are shown in Fig. 2; on the HER2+ and cSCC datasets, the average PCC of Reg2ST is higher than three other methods. Specifically, for the HER2+ dataset, Reg2ST achieved the highest PCC on most of the tissue sections, especially the C1-6 sections, compared with the baseline methods. We note that all methods have low performance on E1-3 and F1-3 tissue sections. HistoGene had the lowest performance in tissue sections. To avoid outliers, we also calculated the median PCC for each method, which is still higher than baseline methods. These results demonstrate the performance of Reg2ST. By recognizing potential patterns in gene expression and applying the learned patterns to histology images, Reg2ST can predict spatial transcriptomics more accurately. It is worth noting that HGGEP performs poorly on the cSCC dataset but performs well on HER2+. This may be due to problems with its parameters. The performance of Loki on HER2+ and cSCC was suboptimal, which may be attributed to several factors. First, the relatively small size of our datasets limits its ability to fully utilize contextual information under the unsupervised inference setting. Second, the large number of target genes in our prediction task may exceed the typical capacity of Loki, which is potentially more suited for predicting a limited set of representative genes, making it difficult to maintain stable performance in a high-dimensional output space.

To investigate whether there is a significant difference between Reg2ST and baselines, the Wilcoxon signed-rank test is performed on PCC results. The experimental results are shown in the Table 2 and Table 3. In the HER2+ dataset, the PCC of Reg2ST was significantly better than that of HisToGene, Hist2ST, THItoGene, and HGGEP, with a corrected significance level of $\alpha = 0.0125$. In the cSCC dataset, the difference between Reg2ST and Hist2ST and THItoGene did not reach significance, which might be due to the small sample size or the higher tissue heterogeneity.

In order to investigate how Reg2ST performs in terms of experimental performance compared with baselines, experiments on computational efficiency are conducted, including the number of parameters, training time, and memory usage of the different models. The specific results are shown in the following Table 4.

Ablation experiment

In this section, our experiment focuses on the effect of each module in the model on the prediction accuracy, including three parts: learning spatial transcriptomics patterns and aligning histology images to spatial transcriptomics, using pretrained large models for histology images feature extraction, and using dynamic GNNs. The results are shown in Fig. 3 and Fig. 4.

The introduction of spatial transcriptomics is removed firstly, including contrastive learning loss to bring spatial transcriptomics features closer to histology image features and feature prediction loss for cross-attention fusion. The mean PCC on the two datasets decreased by 78% and 41%, respectively, indicating that the prediction ability of the model can be effectively improved by introducing spatial transcriptomics. Spatial transcriptomics and histology images seem to be unrelated. However, they are different descriptions of spots, which are just different representations of the same data. By bringing them closer and aligning the images toward genes, it is possible for utilize the potential knowledge to spatial transcriptomics prediction. This results in better prediction accuracy.

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

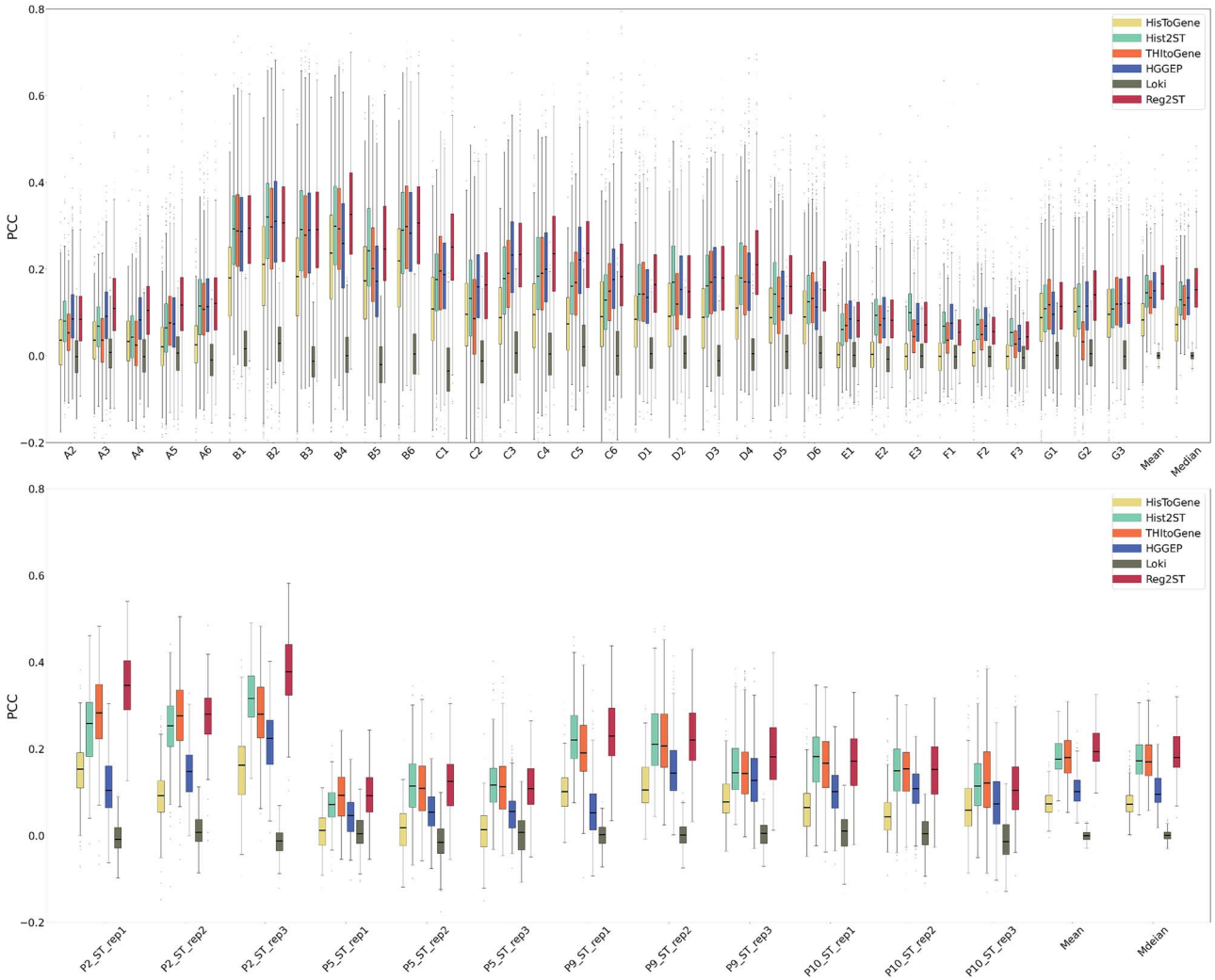

Figure 2. Evaluation of spatial transcriptomics prediction on the HER2+ and cSCC datasets by the PCC between the real and predicted spatial transcriptomics. High-resolution images are available for viewing at https://github.com/Holly-Wang/Reg2ST.

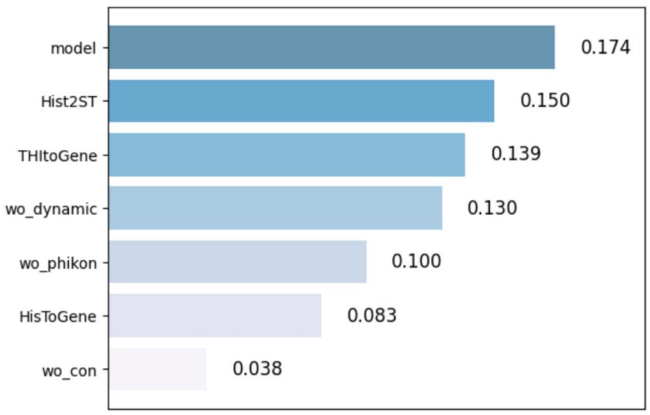

Figure 3. Ablation study on the HER2+ dataset.

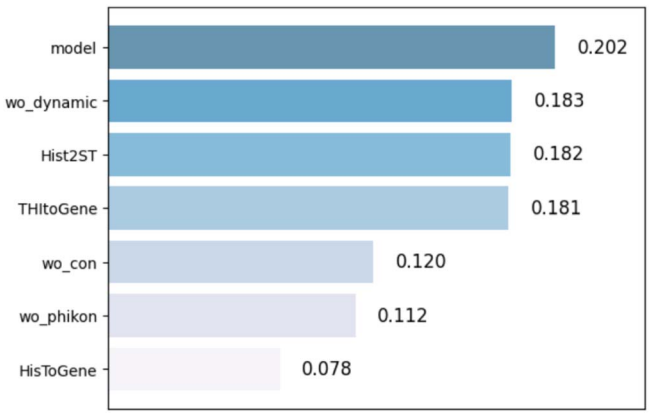

Figure 4. Ablation study on the cSCC dataset.

Then the pretrained model is removed and replaced with ViT for image feature extraction. The result drops by about 43% and 45%, respectively. Batch effects and noise inevitably prevent models from obtaining accurate image features. By using fundamental models pretrained on a wide range of datasets, Reg2ST can obtain features with prior knowledge of biology.

Finally, the dynamic GNN module is removed and replaced with the KNN algorithm to construct the graph, and follow GraphSage [25] method to integrate the critical domain features and global features. Prediction accuracy decreases by 25% and 9%, respectively. Traditional KNN graphs rely only on physical distances and fail to capture functional similarities and organizational hierarchies. Dynamic graphs model deeper interactions by constructing adjacencies through feature similarities. Therefore, ablation experiments illustrate the rationality and effectiveness of Reg2ST.

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

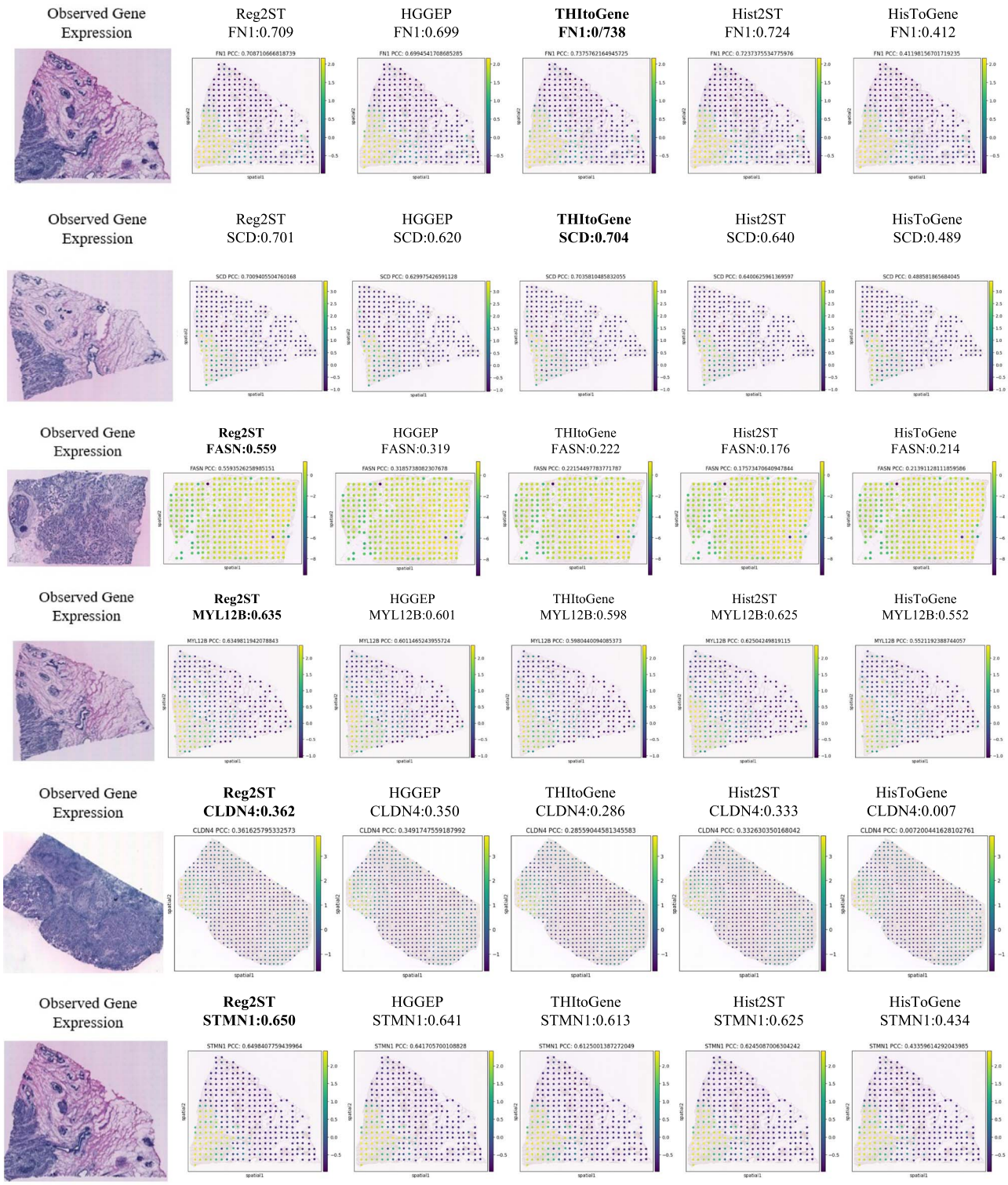

Figure 5. Visualization results on the HER2+ dataset.

Visualization of the predicted gene expression

To further assess predicted gene expression, all genes values of the mean -lg (*P*-value) of all tissue sections are ranked, where the *P*-value of each tissue section was based on predicted and observed gene expression. The top 10 genes with the highest -lgP value are selected for analysis, which are GNAS, FN1, SCD, IGKC, FASN, MYL12B, CLDN4, RHOB, HLA-DRA, and STMN1.

Most of these genes are associated with breast cancer. For example, FN1 encodes a structural protein called fibronectin, a highly conserved macromolecular protein mainly present in the extracellular matrix. It plays a key role in the structure and function of cells. It is a critical multi-regulatory molecule in the tumor stroma group [26]. SCD is of great importance in the *de novo* synthesis of fatty acids, and increased lipid production and

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

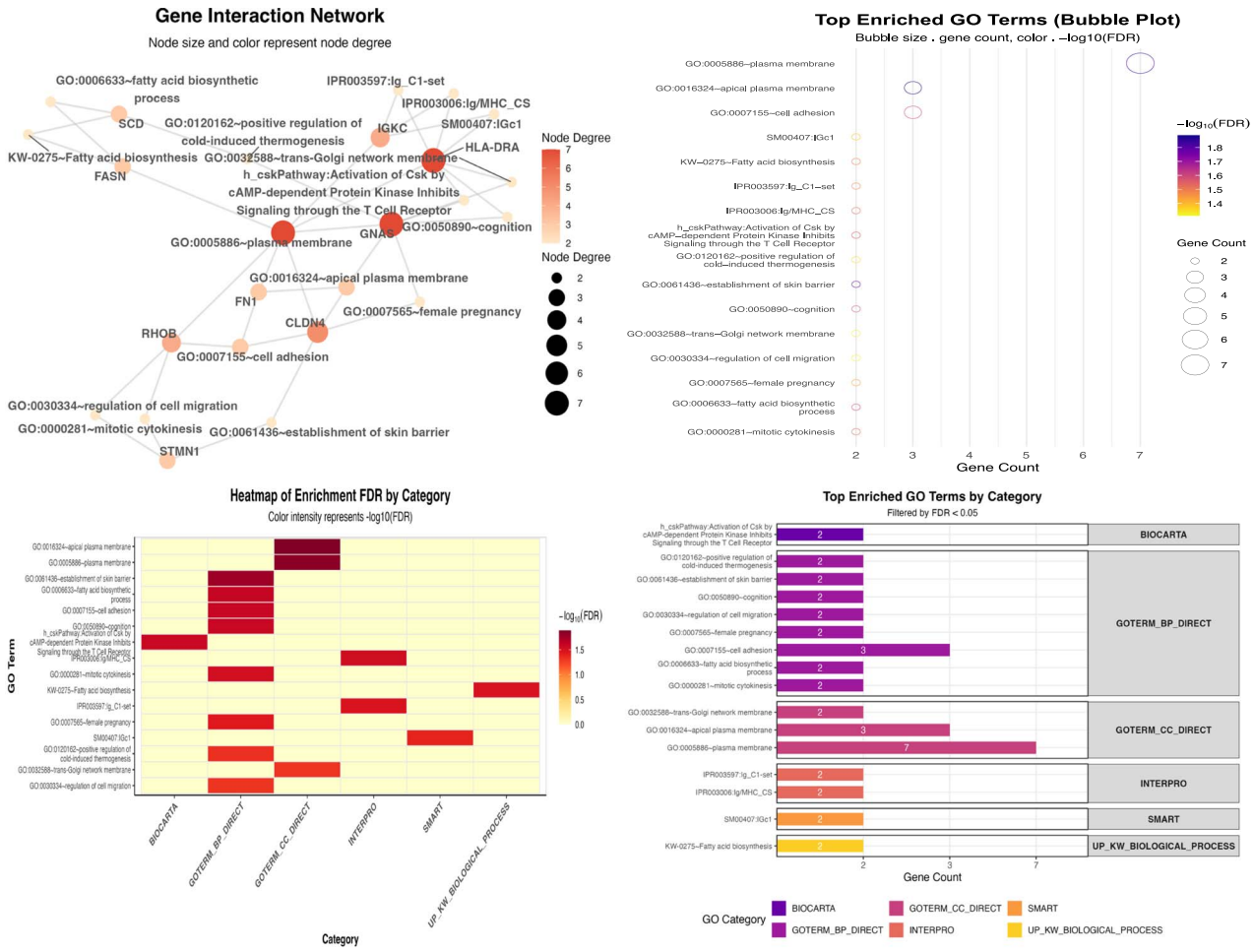

Figure 6. GO analysis and pathway enrichment on HER2+ dataset.

absorption can lead to cancer cell growth and tumorigenesis. In addition, SCD is also necessary in other important cancer-related pathways [27]. FASN encodes a critical enzyme involved in the biosynthesis of lipids in the membrane of proliferating cells. It is still closely associated with the onset and development of tumors [28]. FASN inhibition is a potential therapeutic target for breast cancer by inducing apoptosis of breast cancer cells [29]. A myosin regulatory subunit, enoded by MYL12B, helps regulate the contractile activity of non-muscle cells through phosphorylation. Recently, a study showed that myosin II activity drives tumor development in cancer cells. In contrast, in non-muscle cells, myosin II activation regulates the phosphorylation of light chains [30]. In addition, CLDN4-based cell adhesion signals can accelerate the metabolism and progression of breast cancer through LXR β [31]. STMN1 encodes a phosphoprotein associated with cancer metastasis, which is the most common cause of death in breast cancer patients, proving the association between this gene and breast cancer [32]. It is worth mentioning that studies have shown that the mutation rate of GNAS is very high in cancers such as small intestine cancer, cervical cancer, and colorectal cancer. Although there is no clear study showing that breast cancer is associated with high expression of GNAS, the high expression of GNAS in breast cancer provides a direction for studying the mutant genes of breast cancer [33]. Visualization results are shown in Fig. 5.

Further, GO analysis and pathway enrichment of the above genes are performed. Specifically, we search these genes using the DAVID database to identify GOs and pathways associated with them. The specific information is shown in Fig. 6. Specifically, FASN, SCD act synergistically in the fatty acid synthesis pathway (GO:0006633). Breast cancer cells promote tumor growth and metastasis through enhanced fatty acid synthesis [34]. RHOB and STMN1 drive metastasis through migration regulation (GO:0030334). Targeting the migration GO:0030334 thought pathway is a novel strategy to overcome metastatic resistance in breast cancer [35]. STMN1 and RHOB promote proliferation through regulation of mitosis (GO:0000281). Abnormal mitosis drives genomic instability in breast cancer and drugs targeting this process are used for therapy [36].

Conclusion

In this study, we focus on spatial transcriptomics prediction. The emergence of spatial transcriptomics has made it easier for us to study the growth and differentiation process of cell. It also reveals more profound and clearer biological principles for us. So it is crucial to get accurate and effective spatial transcriptomics. However, spatial transcriptomics sequencing technology costs a lot. To solve the problem, we propose Reg2ST, a deep learning method for spatial transcriptomics prediction. By learning potential patterns from gene expression and aligning histology images with them, Reg2ST obtains better features and more accurate prediction. By doing this, Reg2ST views spatial transcriptomics and histology images as different representations of the same spot. Contrastive

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

learning is used to narrow the distance between them in latent space. To better align histology images with gene expression, Reg2ST used a prediction layer to predict gene features first. Then predicted gene features are used for prediction. To fully exploit the relationships between spots, Reg2ST calculates the similarity between predicted gene features and enhanced image features to construct a graph.We conducted experiments with four baselines on two datasets. Results on PCC, statistical tests, and experimental efficiency proves that Reg2ST outperforms other baselines, which proves the validity and rationality of Reg2ST. Source codes and data are available at https://github.com/Holly-Wang/Reg2ST.

Despite its remarkable prediction performance, we must admit that the datasets we used have some limitations. This is mainly because of sequencing technology and tissue section we chose. On the one hand, both datasets were generated based on 10x Visium technology, whose resolution and gene coverage differ significantly from other technologies, such as Slide-seqV2 [37] and MERFISH [38, 39]. On the other hand, both types of cancers in the dataset we used are solid tumors. It is necessary to add sections of more type of tumors and normal tissue. In addition, only limited spatial transcriptomics datasets and histology images are available, which may lead to overfitting on complex feature extraction models. It is essential to use more accurate models or to employ pretrained large models for feature extraction.

Key Points

- Reg2ST, which is a contrastive-learning-based method that uses gene expression and histology, is proposed for spatial transcriptomics prediction.
- Contrastive learning is used to learn the potential relationships between spatial transcriptomics and histology images and align these modalities, which bridges the gap between visual and transcriptomic data.
- To capture hierarchical and functional relationships between spots, Reg2ST dynamically constructs graphs based on similarity scores derived from modality features.
- Comparison experiments on HER2+ breast cancer and cSCC datasets show that Reg2ST outperforms state-of-the-art methods by significant margins.

Author contributions

Xu Wang (Conceptualization, Formal analysis, Writingoriginal draft), Yaqiu Wang (Formal analysis), Xuan Wang (Formal analysis), Yadong Wang (Formal analysis), and Junyi Li (Conceptualization, Formal analysis, Writingoriginal draft)

Competing interests: No competing interest is declared.

Funding

This work was supported by the grants from the National Key R&D Program of China (2024YFA0919600), and National Natural Science Foundation of China (32470704).

Data availability

Source codes and data are available at https://github.com/Holly-Wang/Reg2ST.

References

1. He B, Bergenstråhle L, Stenbeck L. *et al.* Integrating spatial gene expression and breast tumour morphology via deep learning. *Nat Biomed Eng* 2020;**4**:827–34. https://doi.org/10.1038/s41551-020-0578-x
2. Pang M, Kenong S, Li M. Leveraging information in spatial transcriptomics to predict super-resolution gene expression from histology images in tumors. *BioRxiv* 2021; 2021–11.
3. Zeng Y, Wei Z, Weijiang Y. *et al.* Spatial transcriptomics prediction from histology jointly through transformer and graph neural networks. *Brief Bioinform* 2022;**23**:bbac297.
4. Jia Y, Liu J, Chen L. *et al.* THItoGene: a deep learning method for predicting spatial transcriptomics from histological images. *Brief Bioinform* 2024;**25**:bbad464.
5. Li B, Zhang Y, Wang Q. *et al.* Gene expression prediction from histology images via hypergraph neural networks. *Brief Bioinform* 2024;**25**:bbae500. https://doi.org/10.1093/bib/bbae500
6. Chen W, Pengzhi Zhang TN, Tran YX. *et al.* A visual–omics foundation model to bridge histopathology with spatial transcriptomics. *Nat Methods* 2025;**22**:1–15.
7. Wang Y, Luo J, Jiao S. *et al.* STExplore: an integrated online platform for comprehensive analysis and visualization of spatial transcriptomics data. *Small Methods* 2025;**9**:2401272. https://doi.org/10.1002/smtd.202401272
8. Wang T, Shu H, Jialu H. *et al.* Accurately deciphering spatial domains for spatially resolved transcriptomics with stCluster. *Brief Bioinform* 2024;**25**:bbae329. https://doi.org/10.1093/bib/bbae329
9. Waylen LN, Nim HT, Martelotto LG. *et al.* From whole-mount to single-cell spatial assessment of gene expression in 3D. *Commun Biol* 2020;**3**:602. https://doi.org/10.1038/s42003-020-01341-1
10. Crosetto N, Bienko M, Van Oudenaarden A. Spatially resolved transcriptomics and beyond. *Nat Rev Genet* 2015;**16**:57–66. https://doi.org/10.1038/nrg3832
11. Moor AE, Itzkovitz S. Spatial transcriptomics: paving the way for tissue-level systems biology. *Curr Opin Biotechnol* 2017;**46**: 126–33.
12. Song Q, Jing S. DSTG: deconvoluting spatial transcriptomics data through graph-based artificial intelligence. *Brief Bioinform* 2021;**22**:bbaa414.
13. Schmauch B, Romagnoni A, Pronier E. *et al.* A deep learning model to predict RNA-Seq expression of tumours from whole slide images. *Nat Commun* 2020;**11**:3877. https://doi.org/10.1038/s41467-020-17678-4
14. Dosovitskiy A, Beyer L, Kolesnikov A. *et al.* An image is worth 16x16 words: transformers for image recognition at scale. arXiv preprint arXiv:2010.11929. 2020.
15. Vaswani A, Shazeer N, Parmar N. *et al.* Attention is all you need. *Advances in neural information processing systems* 2017;**30**.
16. Keyulu X, Li C, Tian Y. *et al.* Representation learning on graphs with jumping knowledge networks. In:*International Conference on Machine Learning*, pp. 5453–62. PMLR, 2018.
17. Kipf TN, Welling M. Semi-supervised classification with graph convolutional networks. arXiv preprint arXiv:1609.02907. 2016.
18. Mazzia V, Salvetti F, Chiaberge M. Efficient-CapsnNet: capsule network with self-attention routing. *Sci Rep* 2021;**11**:14634. https://doi.org/10.1038/s41598-021-93977-0
19. Veličković P, Cucurull G, Casanova A. *et al.* Graph attention networks. arXiv preprint arXiv:1710.10903. 2017.

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

20. Eraslan G, Simon LM, Mircea M. *et al.* Single-cell RNA-seq denoising using a deep count autoencoder. *Nat Commun* 2019;**10**:390.
21. Andersson A, Larsson L, Stenbeck L. *et al.* Spatial deconvolution of HER2-positive breast cancer delineates tumor-associated cell type interactions. *Nat Commun* 2021;**12**:6012.
22. Ji AL, Rubin AJ, Thrane K. *et al.* Multimodal analysis of composition and spatial architecture in human squamous cell carcinoma. *Cell* 2020;**182**:497–514.e22. https://doi.org/10.1016/j.cell.2020.05.039
23. Filiot A, Jacob P, Kain AM. *et al.* Phikon-v2, a large and public feature extractor for biomarker prediction. arXiv preprint arXiv:2409.09173. 2024.
24. Li J, Chen Y, Chu H. *et al.* Dynamic graph representation with knowledge-aware attention for histopathology whole slide image analysis. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11323–32, 2024.
25. Hamilton W, Ying Z, Leskovec J. Inductive representation learning on large graphs. *Advances in neural information processing systems* 2017;**30**.
26. Spada S, Tocci A, Di Modugno F. *et al.* Fibronectin as a multiregulatory molecule crucial in tumor matrisome: from structural and functional features to clinical practice in oncology. *J Exp Clin Cancer Res* 2021;**40**:102. https://doi.org/10.1186/s13046-021-01908-8
27. Sen U, Coleman C, Sen T. Stearoyl coenzyme A desaturase-1: multitasker in cancer, metabolism, and ferroptosis. *Trends Cancer* 2023;**9**:480–9. https://doi.org/10.1016/j.trecan.2023.03.003
28. Flavin R, Peluso S, Nguyen PL. *et al.* Fatty acid synthase as a potential therapeutic target in cancer. *Future Oncol* 2010;**6**:551–62. https://doi.org/10.2217/fon.10.11
29. Menendez JA, Lupu R. Fatty acid synthase (FASN) as a therapeutic target in breast cancer. *Expert Opin Ther Targets* 2017;**21**:1001–16. https://doi.org/10.1080/14728222.2017.1381087
30. Georgouli M, Herraiz C, Crosas-Molist E. *et al.* Regional activation of myosin II in cancer cells drives tumor progression via a secretory cross-talk with the immune microenvironment. *Cell* 2019;**176**:757–774.e23. https://doi.org/10.1016/j.cell.2018.12.038
31. Murakami-Nishimagi Y, Sugimoto K, Kobayashi M. *et al.* Claudin-4-adhesion signaling drives breast cancer metabolism and progression via liver X receptor β. *Breast Cancer Res* 2023;**25**:41. https://doi.org/10.1186/s13058-023-01646-z
32. Kuang X-Y, Jiang H-S, Li K. *et al.* The phosphorylation-specific association of STMN1 with GRP78 promotes breast cancer metastasis. *Cancer Lett* 2016;**377**:87–96. https://doi.org/10.1016/j.canlet.2016.04.035
33. Johannet P, Abdelfattah S, Wilde C. *et al.* Molecular and clinicopathologic impact of GNAS variants across solid tumors. *J Clin Oncol* 2024;**42**:3847–57. https://doi.org/10.1200/JCO.24.00186
34. Menendez JA, Lupu R. Fatty acid synthase and the lipogenic phenotype in cancer pathogenesis. *Nat Rev Cancer* 2007;**7**:763–77.
35. Li D-M, Feng Y-M. Signaling mechanism of cell adhesion molecules in breast cancer metastasis: potential therapeutic targets. *Breast Cancer Res Treat* 2011;**128**:7–21. https://doi.org/10.1007/s10549-011-1499-x
36. Duijf PHG, Nanayakkara D, Nones K. *et al.* Mechanisms of genomic instability in breast cancer. *Trends Mol Med* 2019;**25**:595–611. https://doi.org/10.1016/j.molmed.2019.04.004
37. Stickels RR, Murray E, Kumar P. *et al.* Highly sensitive spatial transcriptomics at near-cellular resolution with Slide-seqv2. *Nat Biotechnol* 2021;**39**:313–9. https://doi.org/10.1038/s41587-020-0739-1
38. Moffitt JR, Bambah-Mukku D, Eichhorn SW. *et al.* Molecular, spatial, and functional single-cell profiling of the hypothalamic preoptic region. *Science* 2018;**362**:792–792. https://doi.org/10.1126/science.aau5324
39. Chen KH, Boettiger AN, Moffitt JR. *et al.* Spatially resolved, highly multiplexed RNA profiling in single cells. *Science* 2015;**348**:aaa6090.

Downloaded from https://academic.oup.com/bib/article/26/4/bbaf425/8237539 by guest on 26 March 2026

© The Author(s) 2025. Published by Oxford University Press. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial License (https://creativecommons.org/licenses/by-nc/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.
Briefings in Bioinformatics, 2025, **26(4)**, bbaf425
https://doi.org/10.1093/bib/bbaf425
Problem Solving Protocol